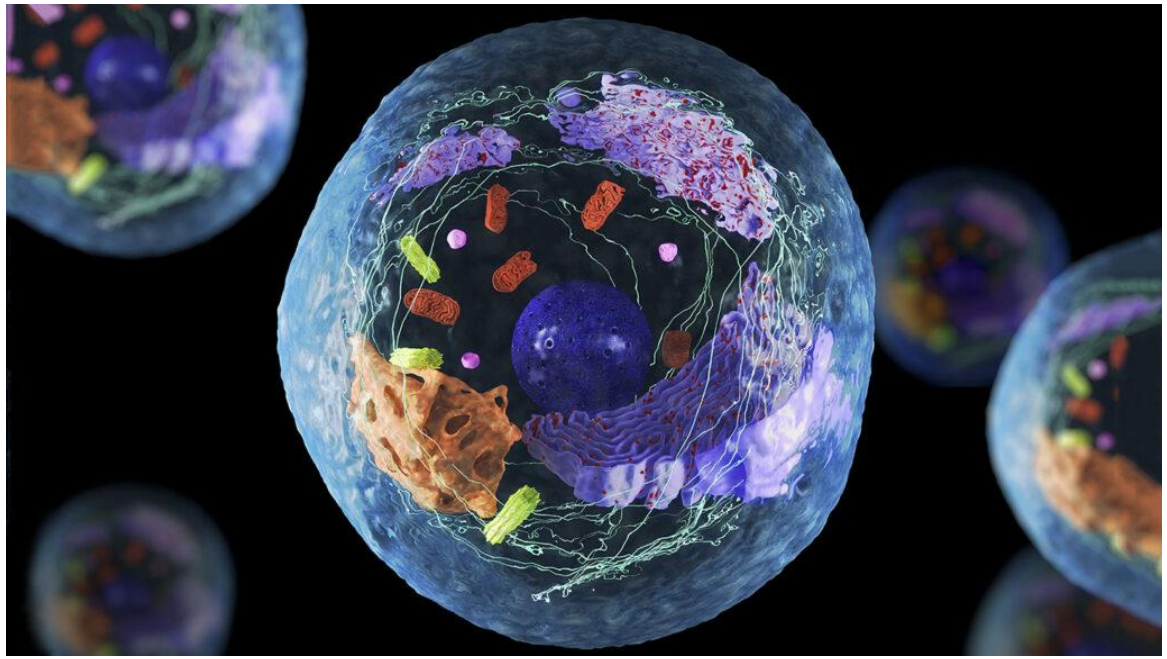
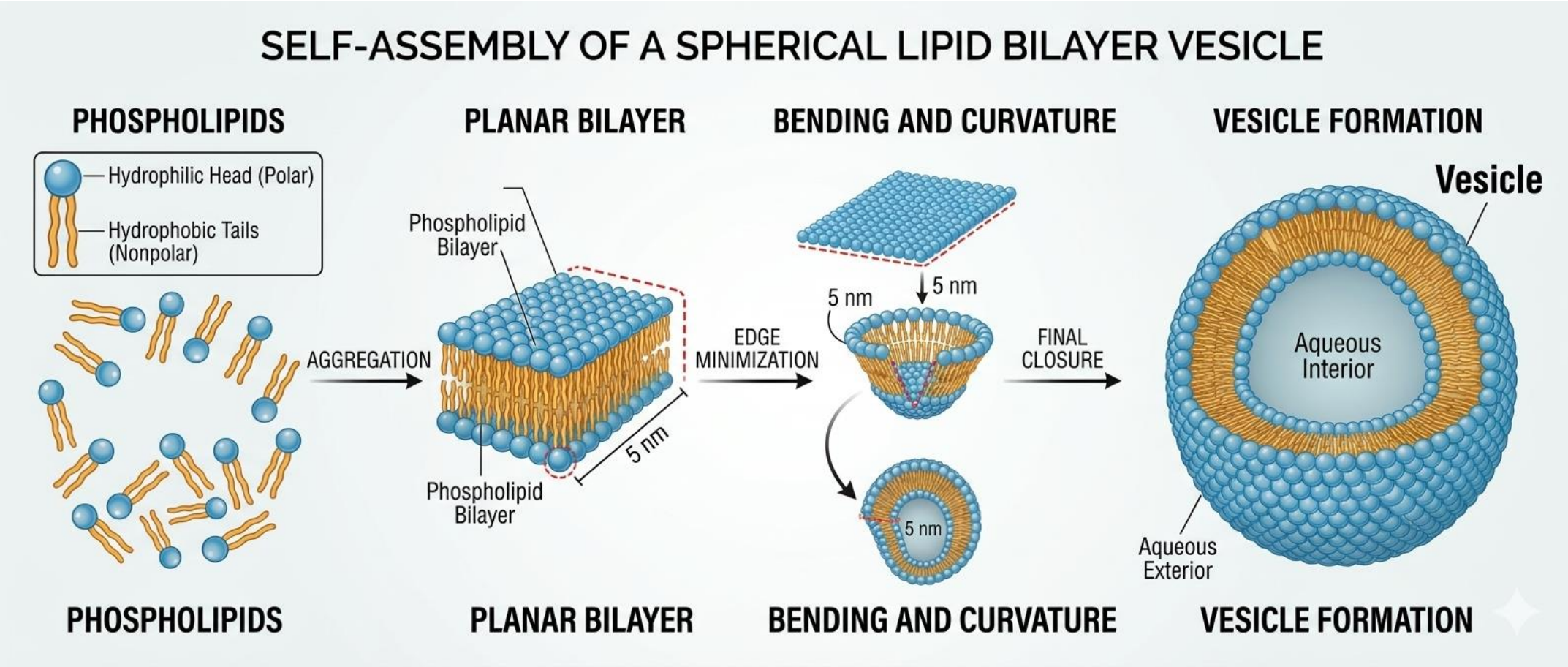
Cell Membrane

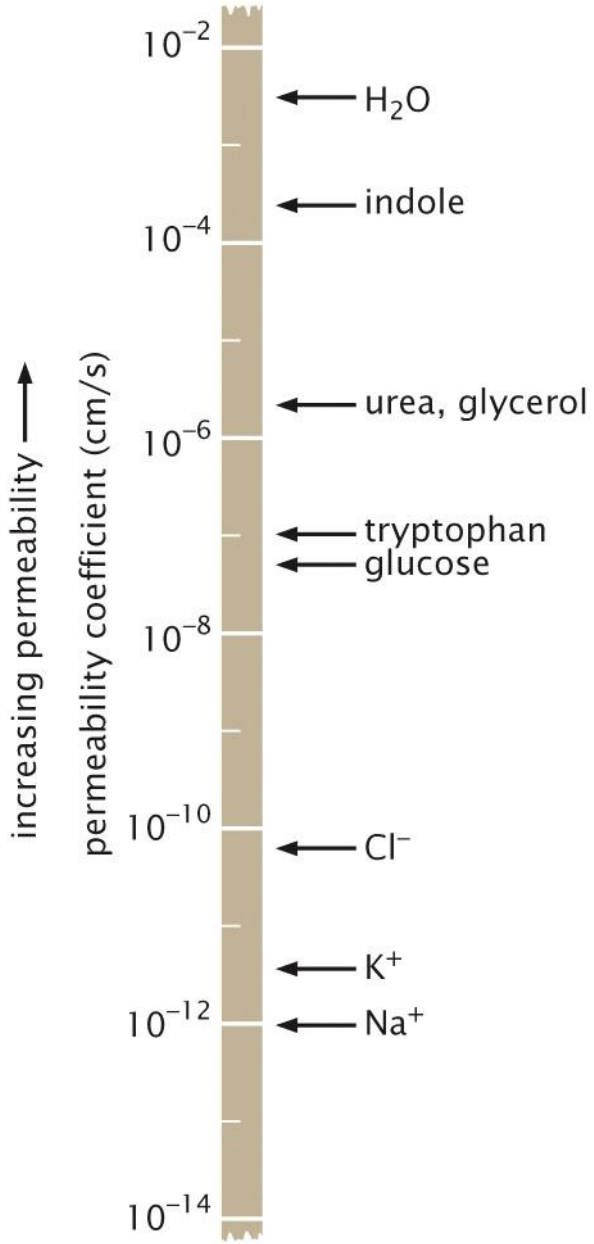
Lecture 14



Membrane formation



Membrane transport



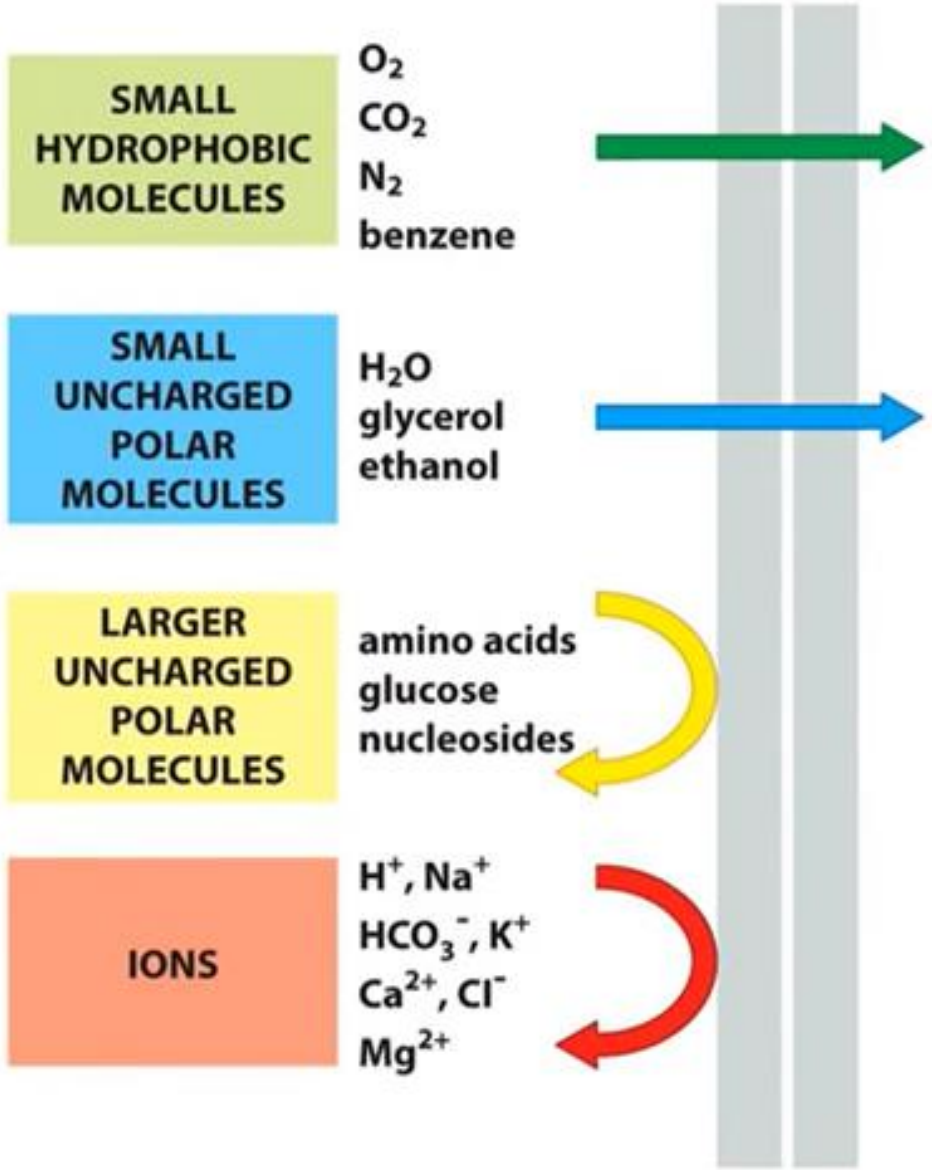


Figure 11.11 Physical Biology of the Cell, 2ed. (© Garland Science 2013)

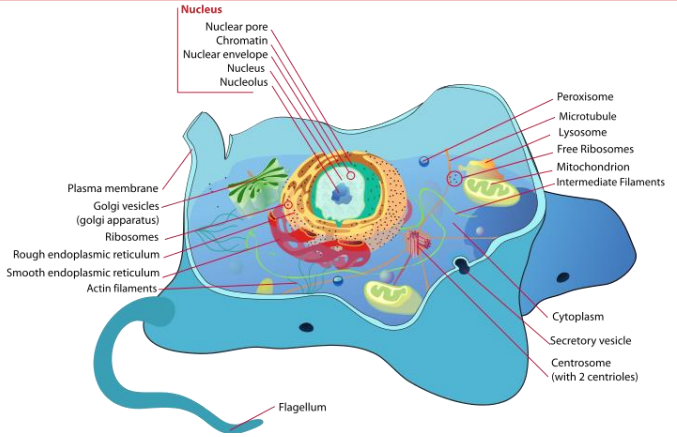
Membrane transport

Particle	Permeability Coefficient (P)	Pure Bilayer Transit Time
Water Molecule ($\mathbf{H_2O}$)	5×10^{-5} m/s	**0.1 ms**
Potassium Ion ($\mathbf{K^+}$)	1×10^{-14} m/s	5.8 Days (500,000 s)

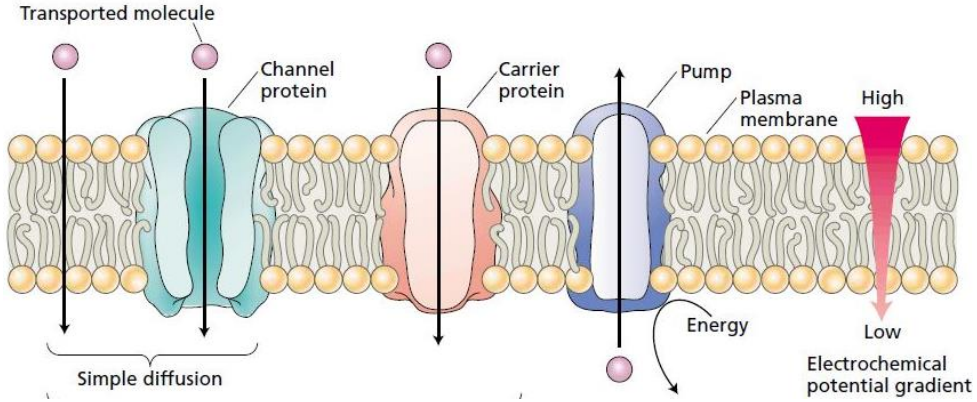
A water molecule zips across the pure lipid interior **5 billion times faster** than a potassium ion simply because it lacks a net charge

Importance of cell (Plasma) membrane

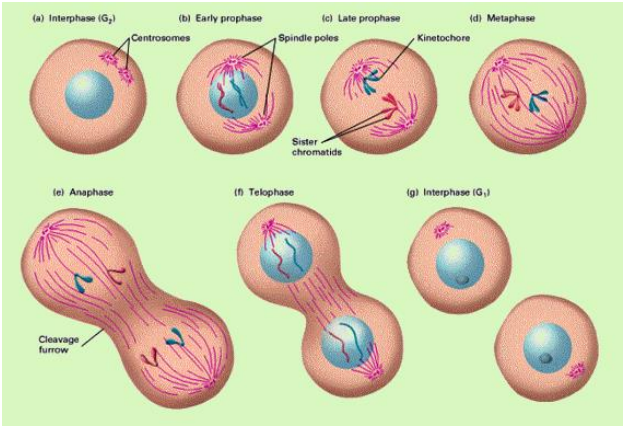
- Separate cellular contents from the external world



- Must allow passage of critical nutrients and waste products.

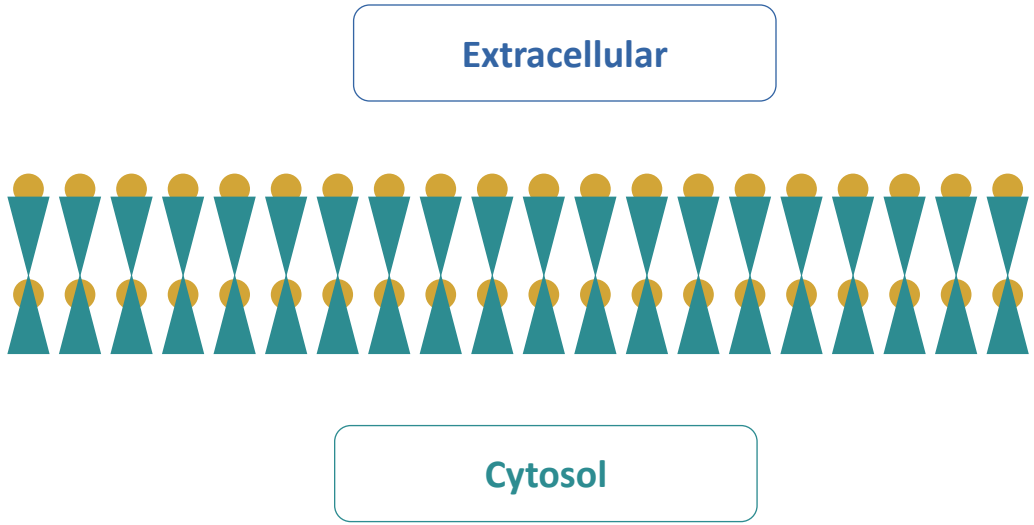


- Must be physically flexible



What is a cell membrane?

- Thin, dynamic boundary separating cytoplasm from the extracellular environment.
- Main components: phospholipids, proteins, cholesterol and carbohydrates.
- Selective permeability lets the cell control its internal chemical environment.
- Membranes are fluid and dynamically reorganized.



Mosaic model



Dynamic processes within lipid bilayers

diffusion of lipids
(rapid)

diffusion of membrane proteins
(slower, react to membrane tension)

flipping of lipids
(extremely slow)

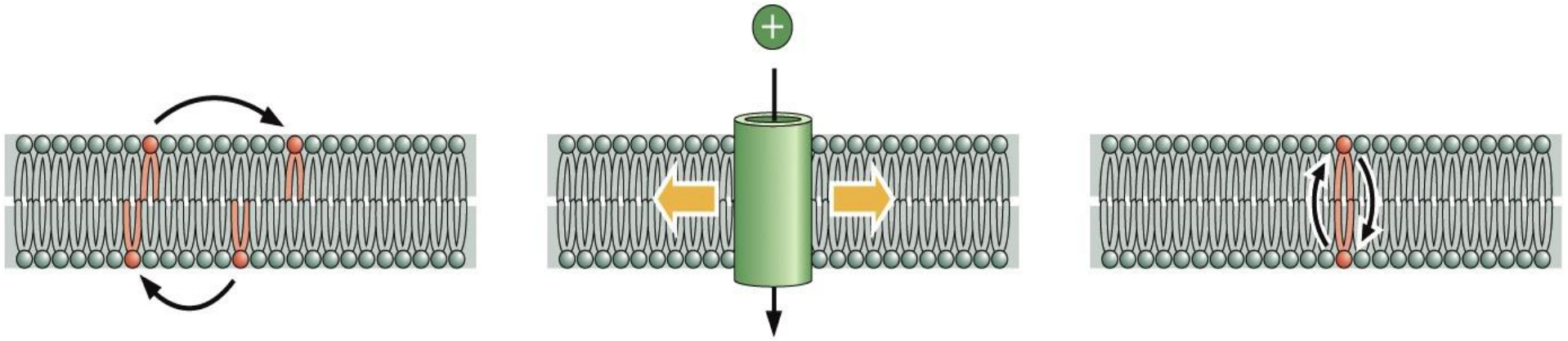
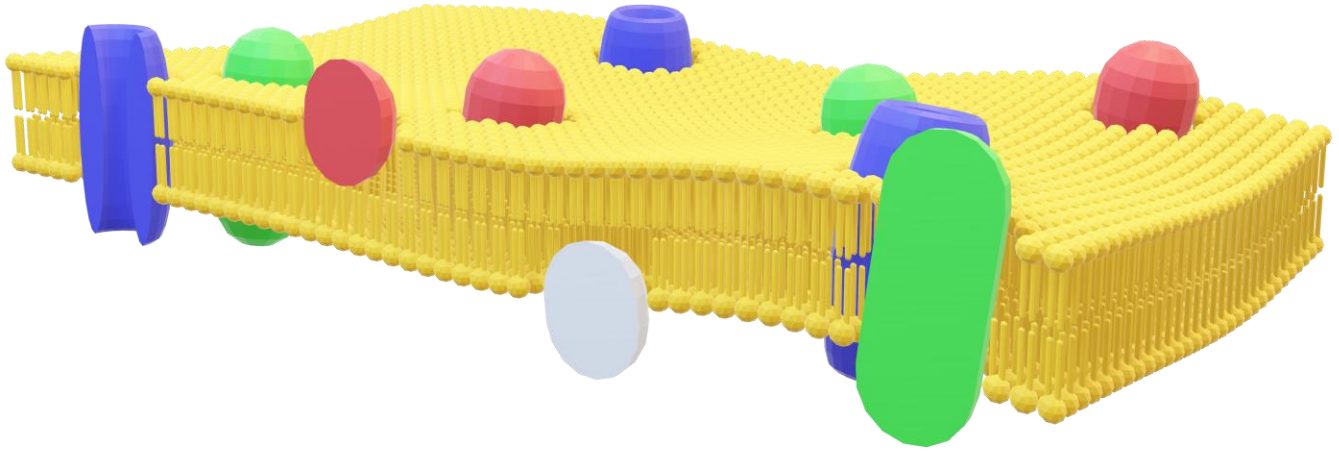


Figure 11.1b Physical Biology of the Cell, 2ed. (© Garland Science 2013)

That is why this structure is also known as “Fluid-Mosaic model”

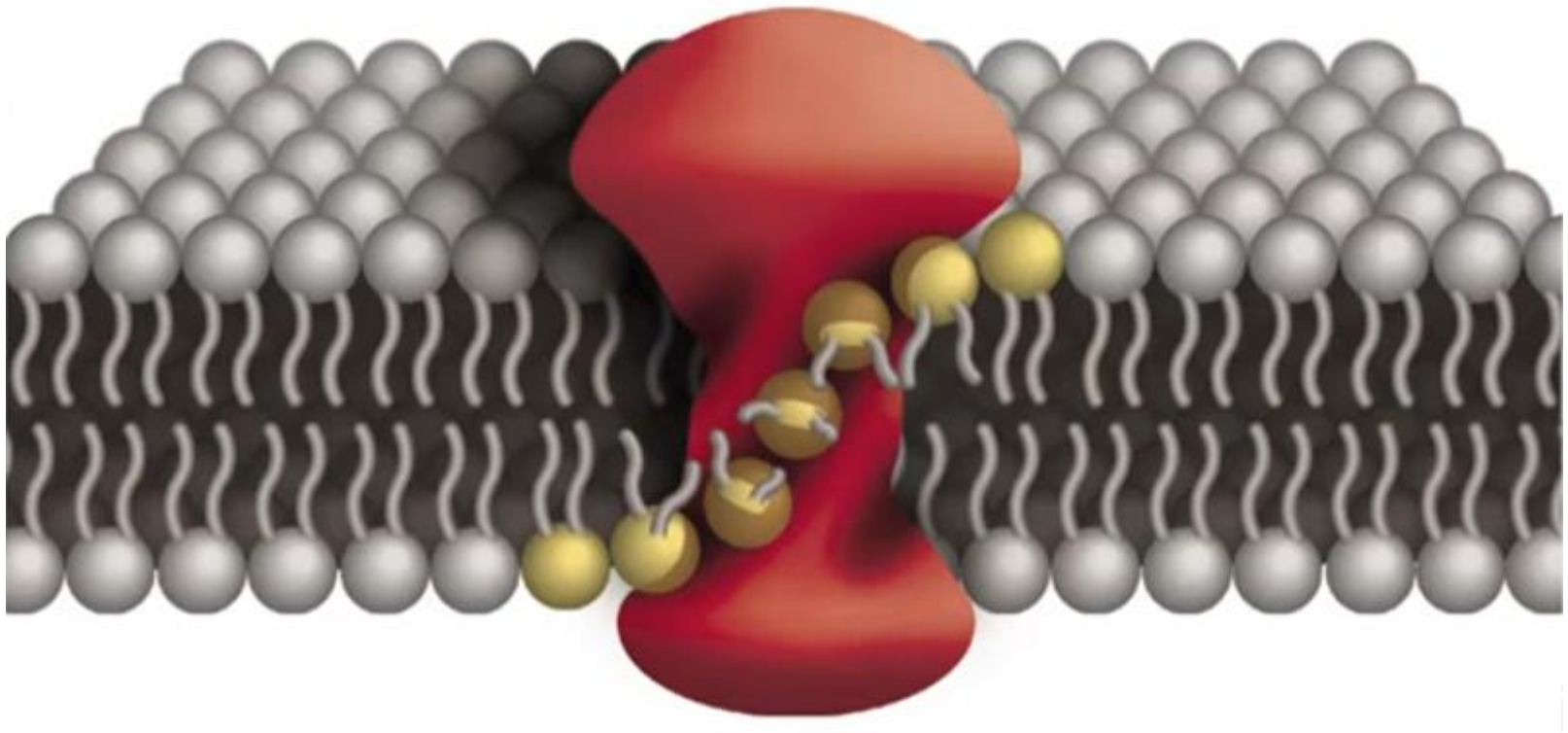
The fluid mosaic model

- Phospholipids form the continuous matrix.
- Membrane proteins perform most specialized transport and signaling functions.
- Cholesterol modifies packing, fluidity and permeability.
- Carbohydrates on the extracellular surface contribute to recognition and adhesion.



https://sketchfab.com/

Dynamic processes within lipid bilayers



Types of Cellular Transport

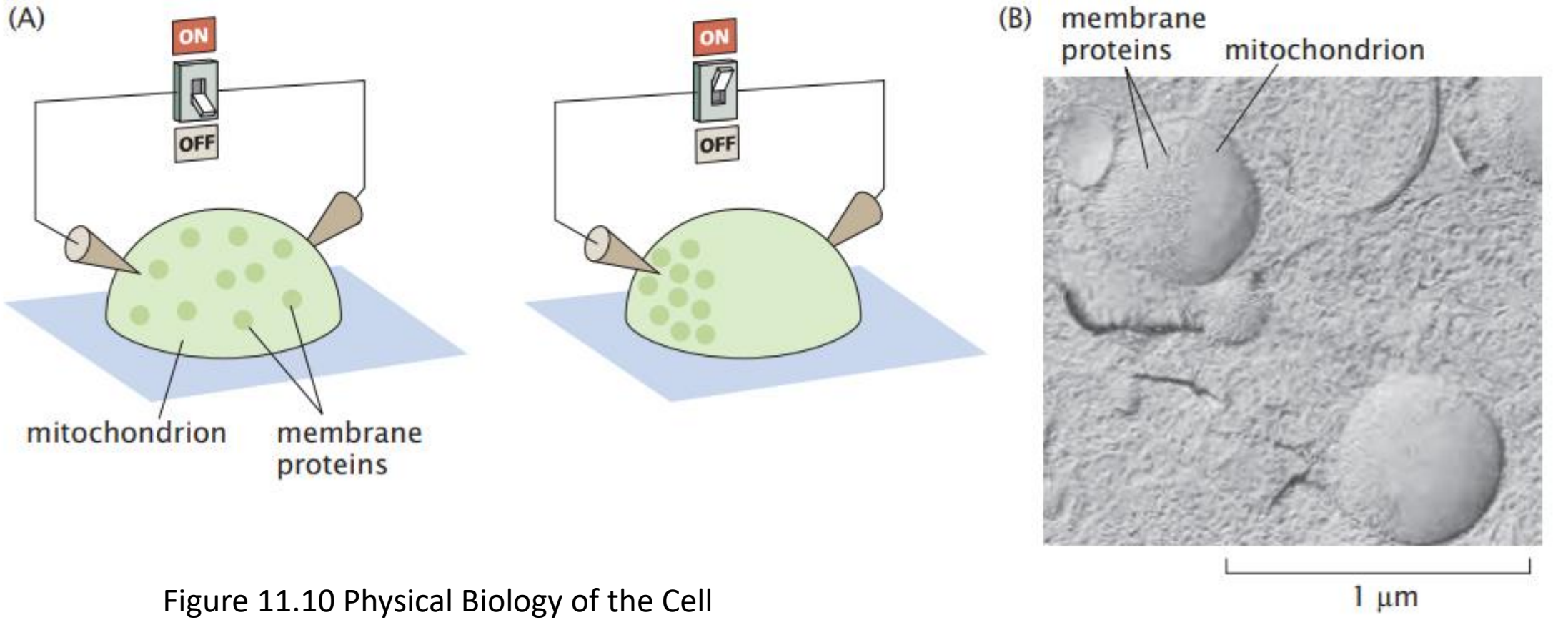


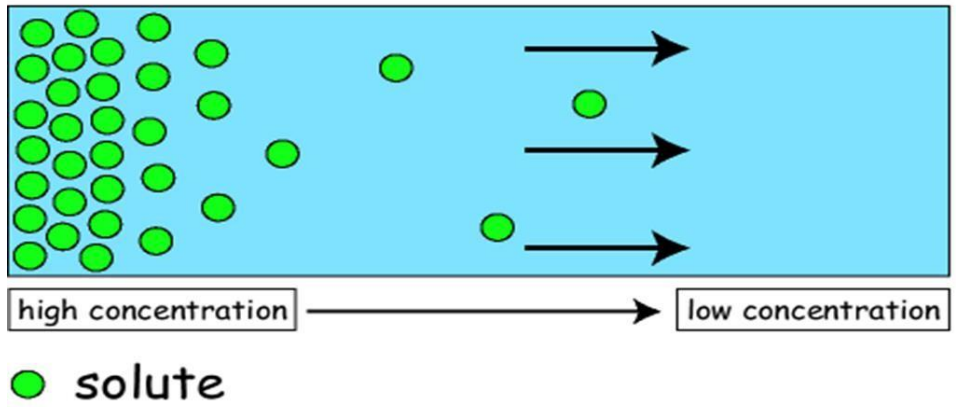
Figure 11.10 Physical Biology of the Cell

Membrane characteristics

- Self-assembly
- Self-healing
- Selective Permeability
- Fluidity
- Asymmetry
- Electrical Polarization (Membrane Potential)
- High flexibility

Diffusion & Osmosis

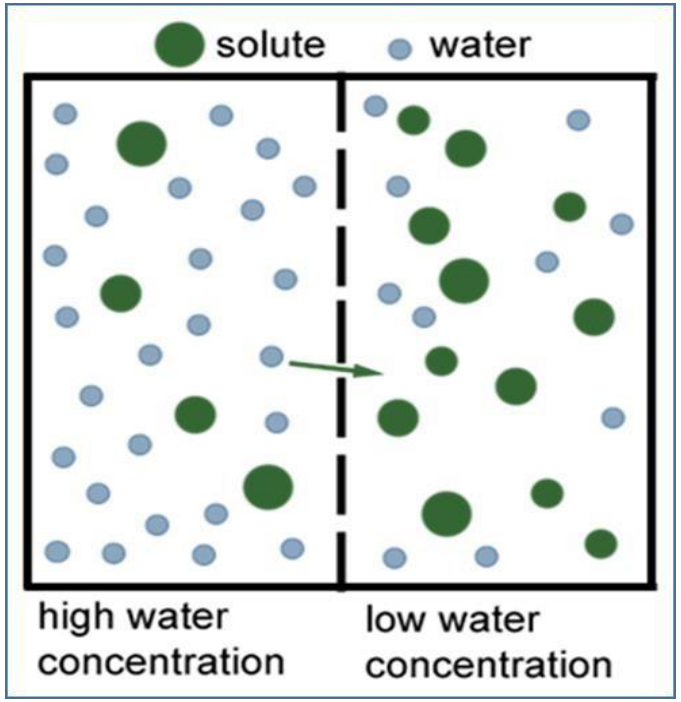
Diffusion



Solute moves DOWN the concentration gradient.

HIGH to LOW

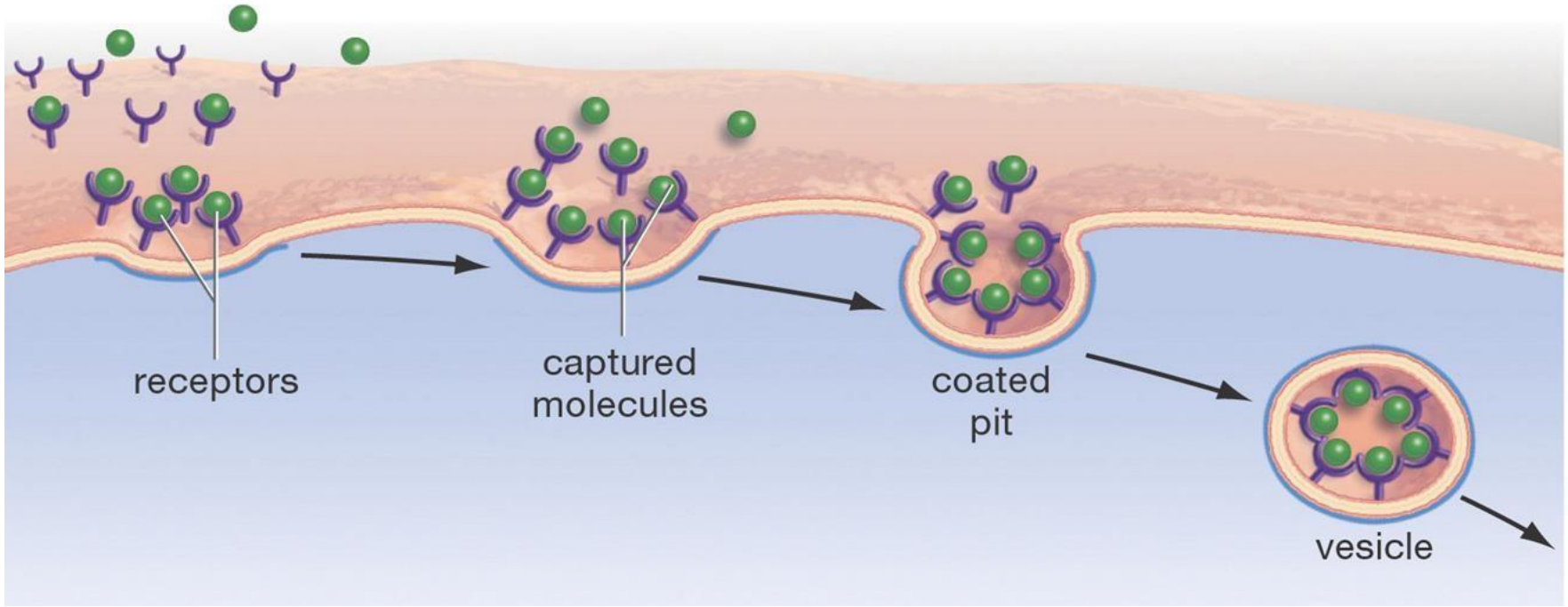
Osmosis



Movement of water across a membrane.

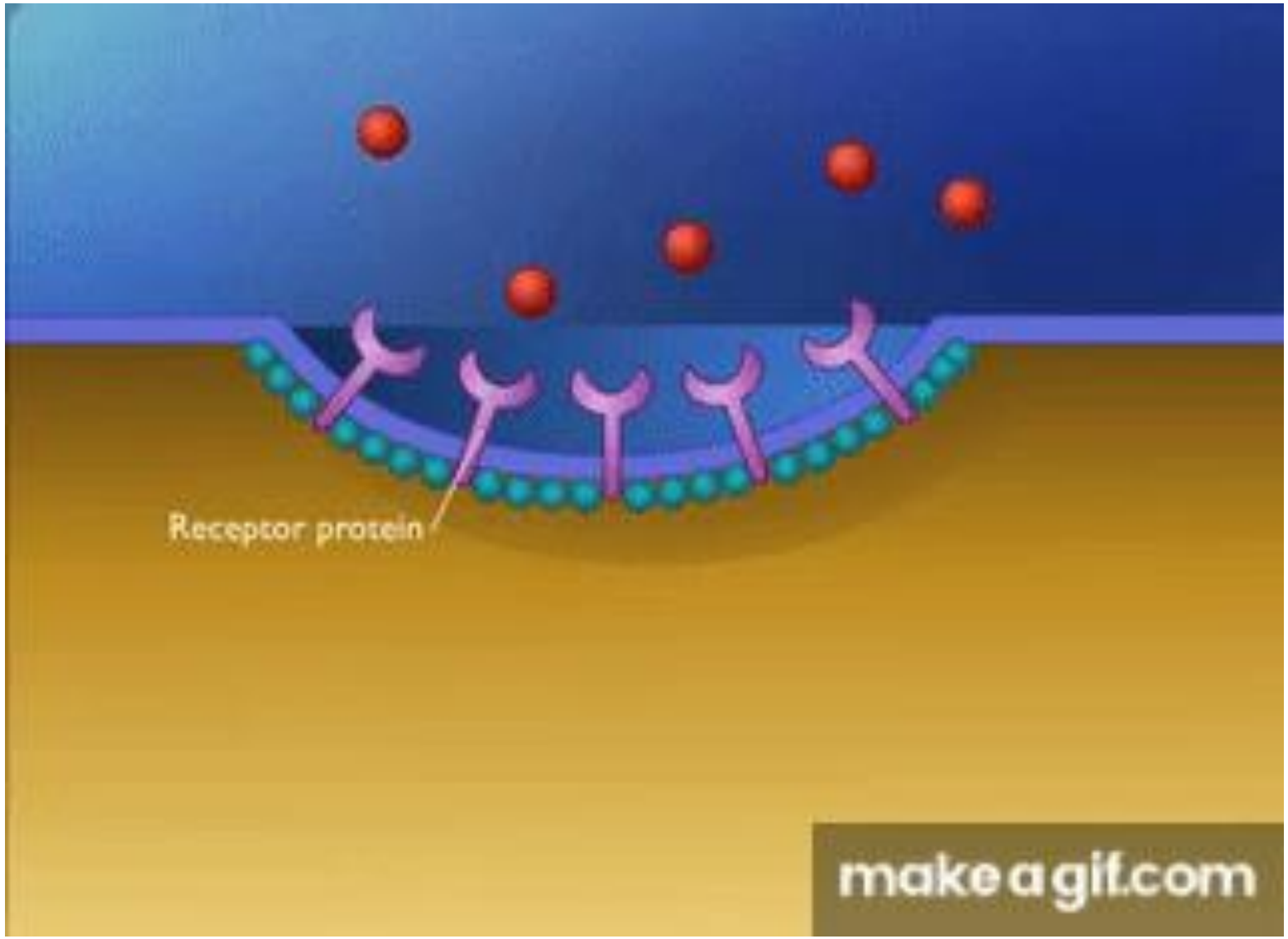
Moves from HIGH water concentration to LOW water concentration.

Receptor-Mediated Endocytosis

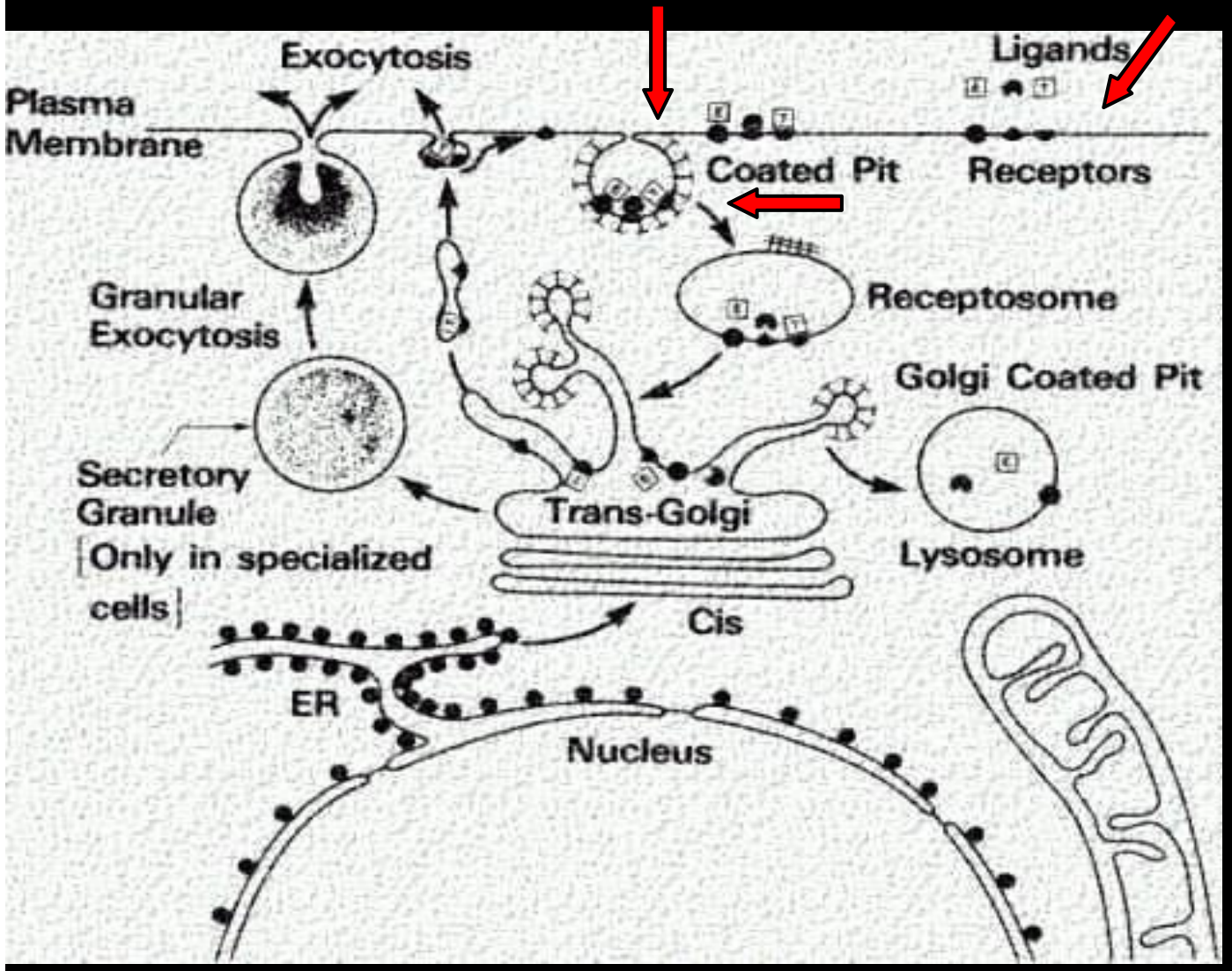


Some integral proteins have receptors on their surface to recognize & take in hormones, growth factors, cholesterol, etc.

Endocytosis

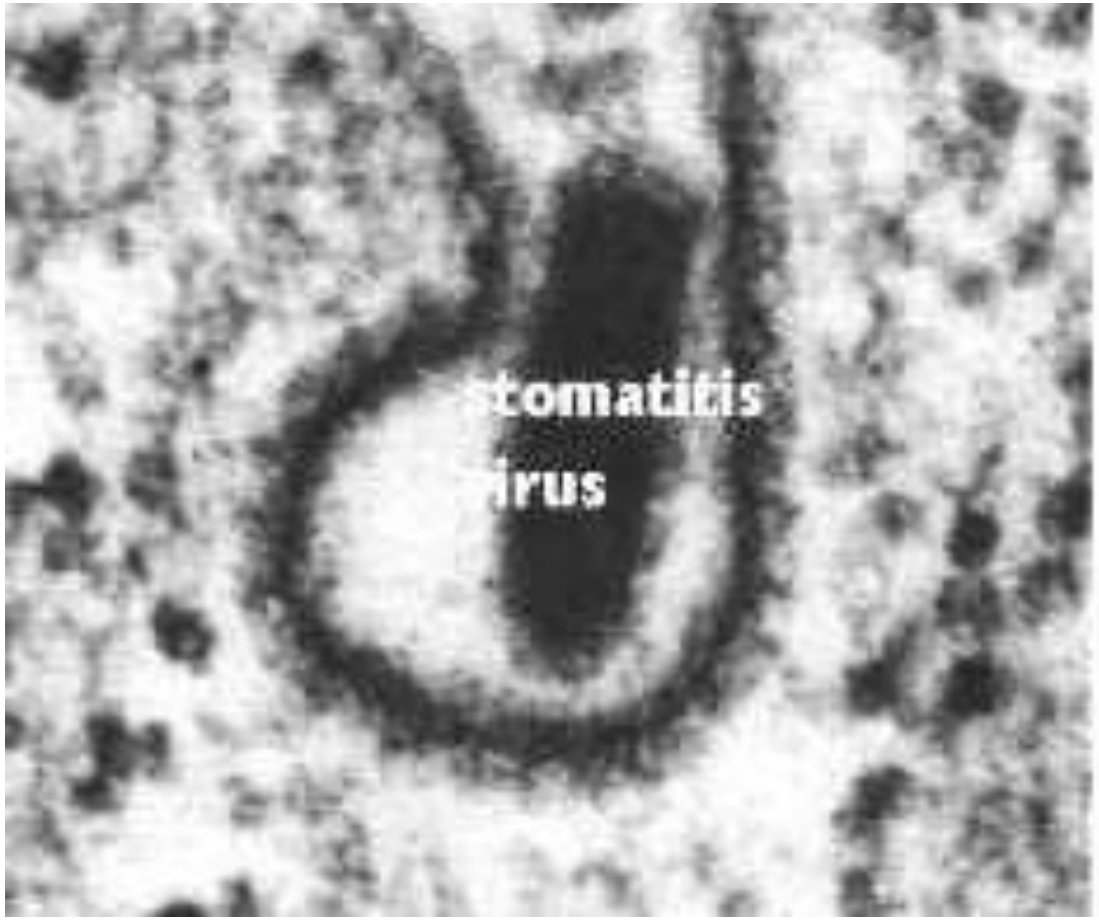


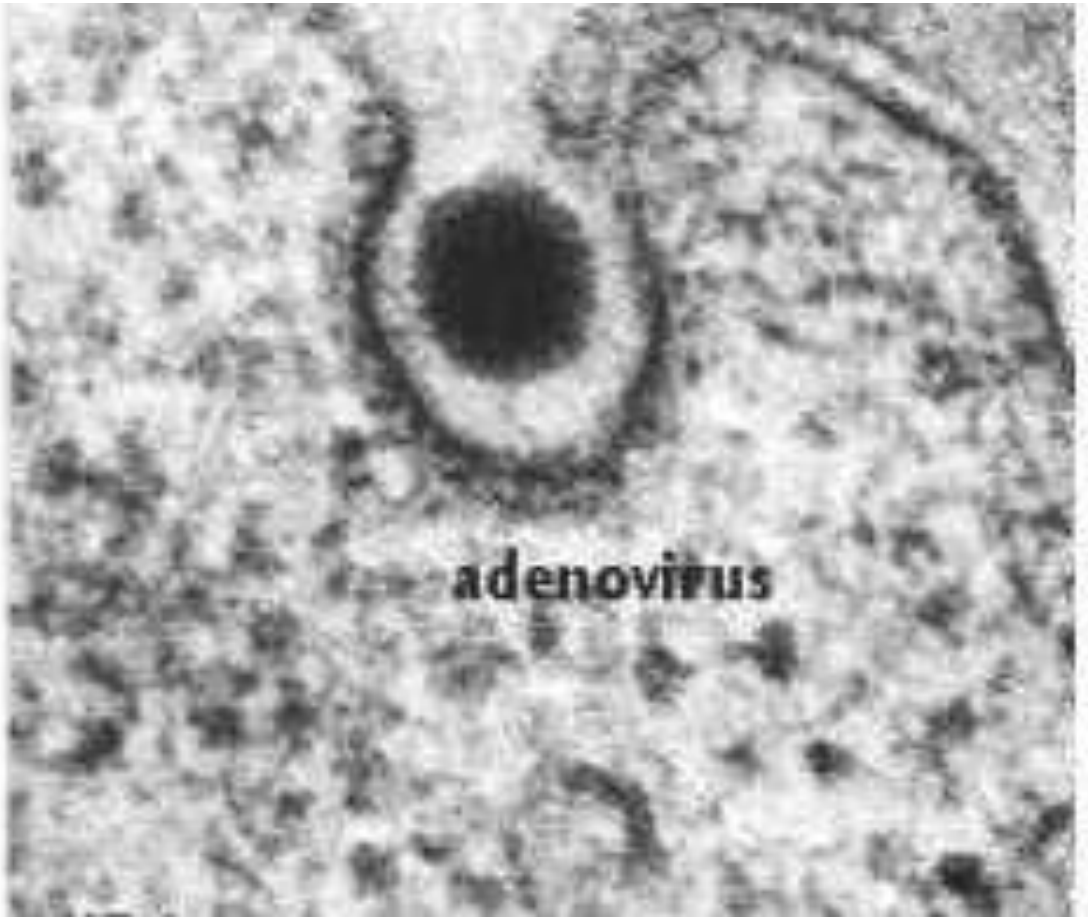
Endocytosis



- Ligands bind receptors
- Receptor/ Ligand complexes cluster, (capping)
- Receptor/ ligand is brought in via clathrin coated pit.

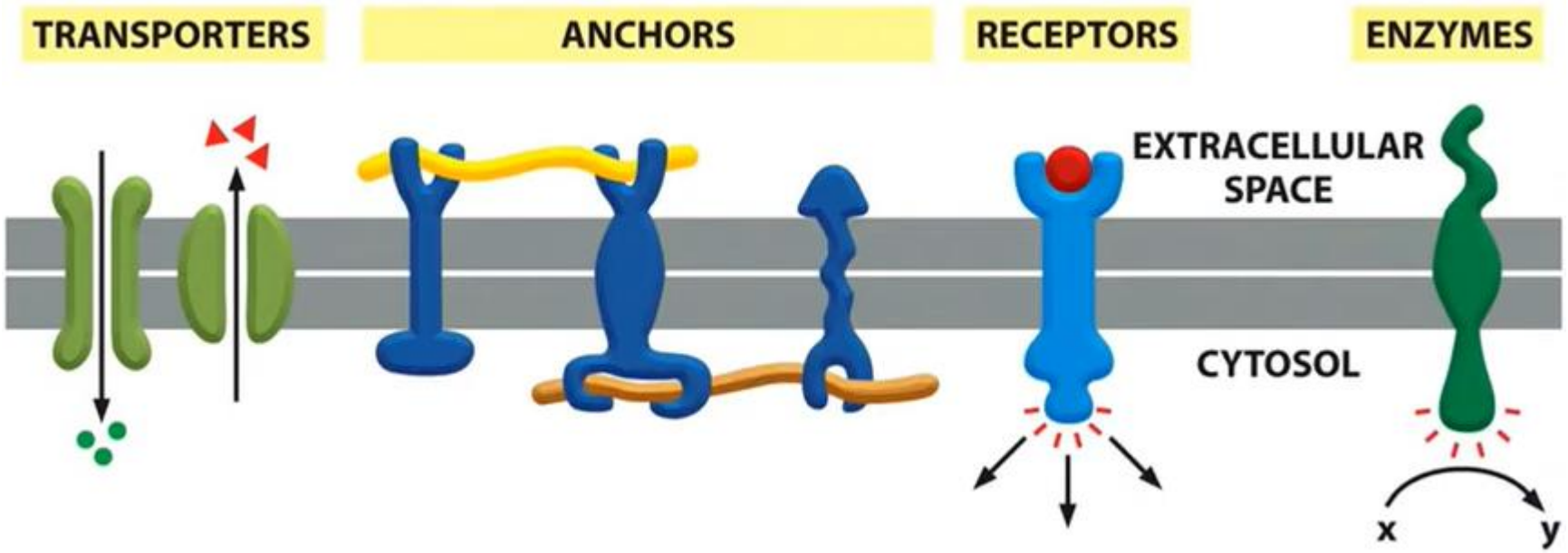
Entry of different viruses in the cell





Membrane proteins

- The lipids of the membrane are largely involved in structural integrity of cell.
- Lipids are not functional, and hence can't do much activity.
- The activity and function of the cell membrane is carried out by membrane proteins



Most common function of the membrane proteins is transport.

Membrane proteins: major functional classes

- Channels — selective hydrophilic pathways for ions or water.
- Pumps — use energy to establish or maintain gradients.
- Receptors — detect extracellular signals and initiate intracellular responses.
- Enzymes, adhesion proteins and anchors — perform additional membrane functions.

Membrane transport: three basic mechanisms

- Simple diffusion: movement down a concentration gradient without a protein.
- Facilitated diffusion: protein-assisted movement down an electrochemical gradient.
- Active transport: energy-coupled movement that can occur against a gradient.
- For ions, both concentration and electrical gradients contribute to the driving force.

Types of Cellular Transport

Passive Transport

Cell doesn't use energy

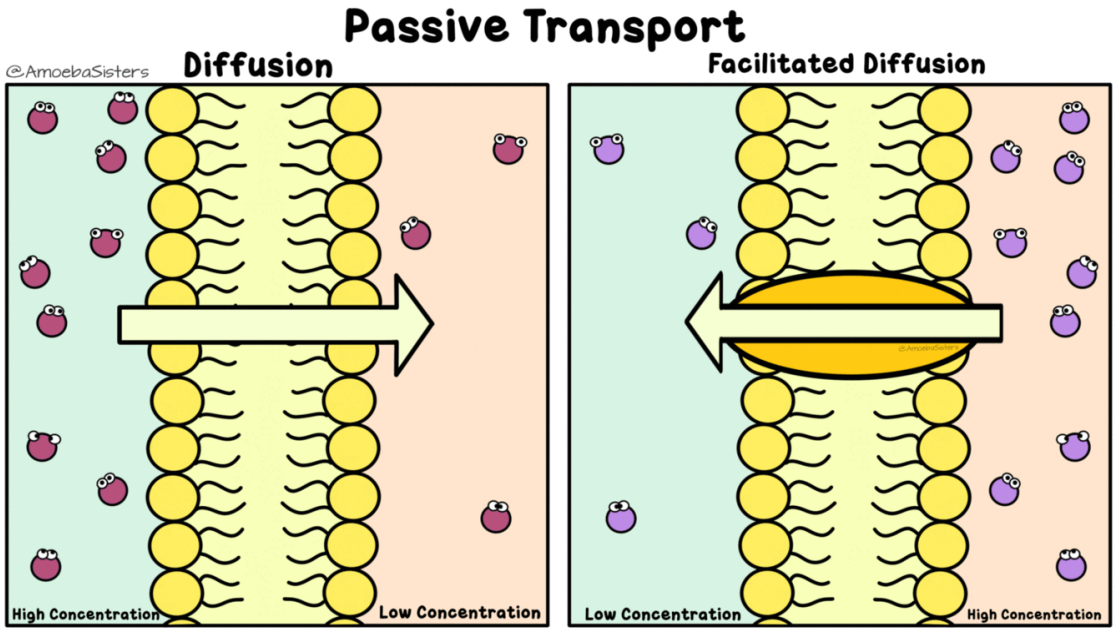
1. Diffusion
2. Facilitated Diffusion
3. Osmosis

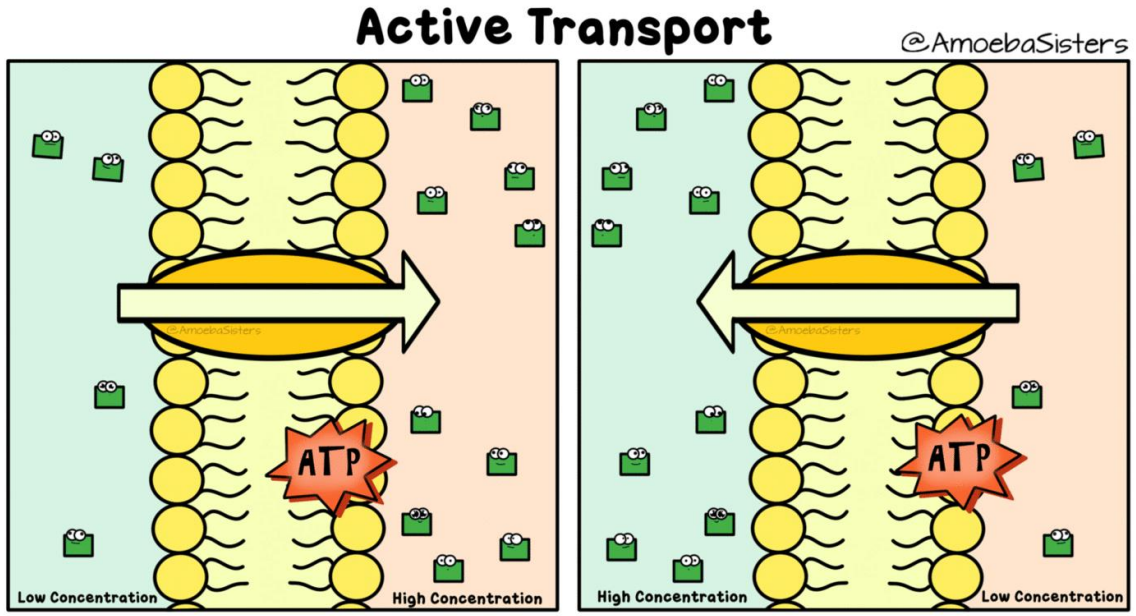
Active Transport

Cell does use energy

1. Protein Pumps
2. Endocytosis
3. Exocytosis

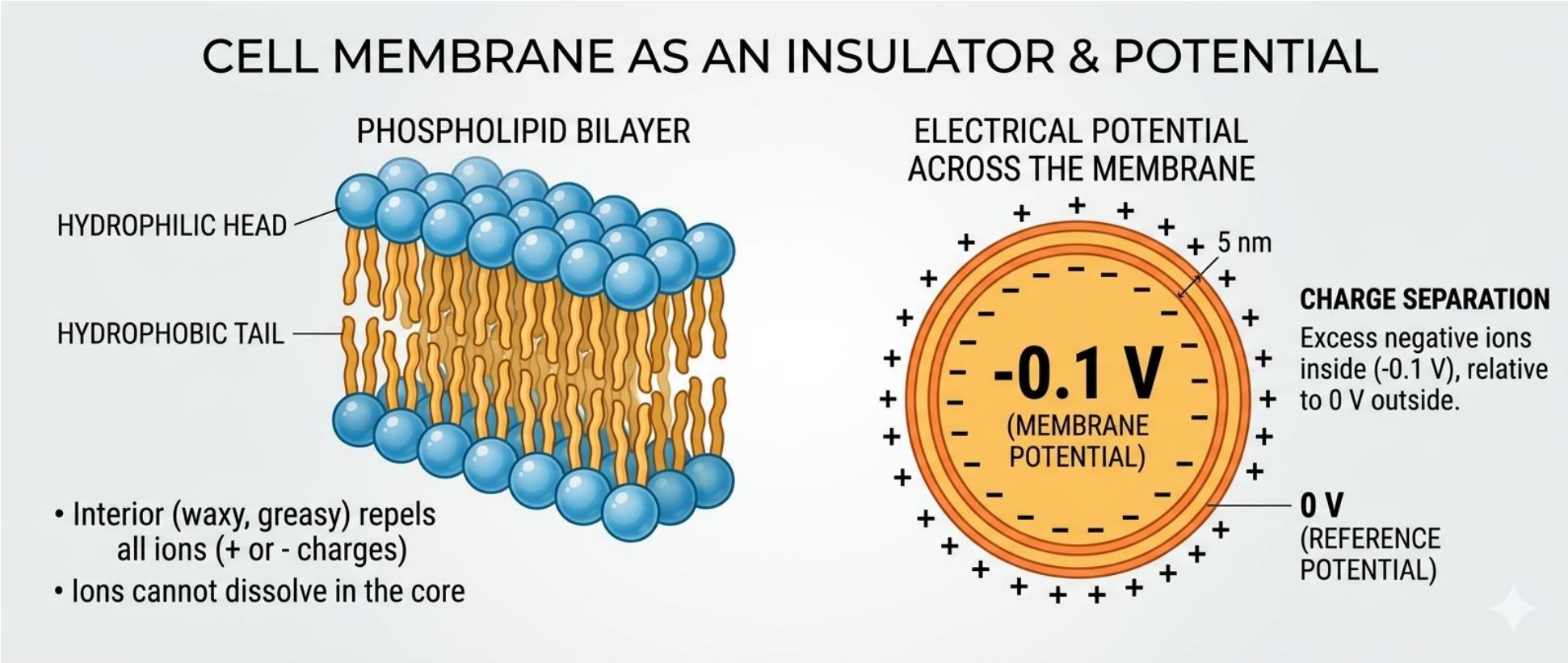
Types of Cellular Transport



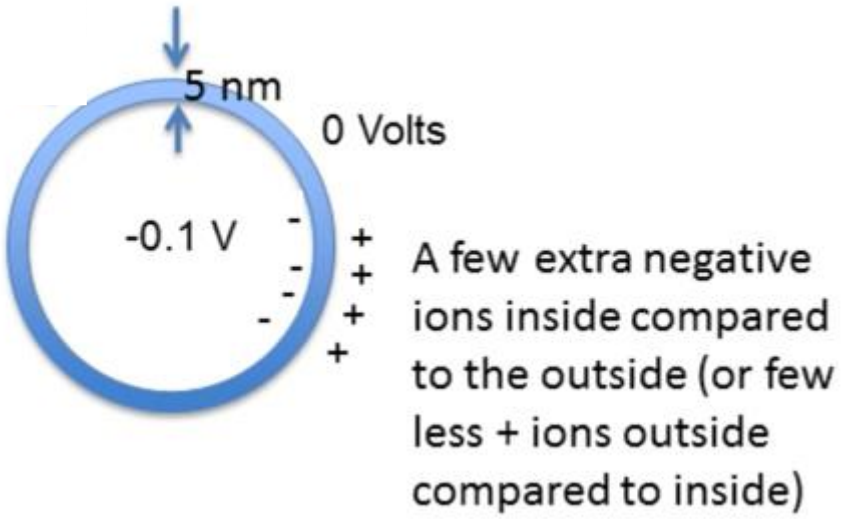


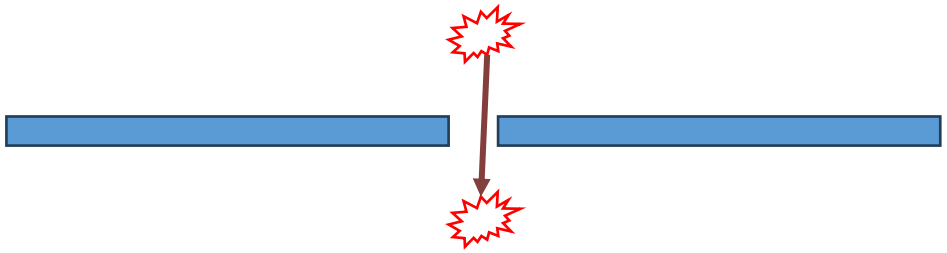
https://www.amoebasisters.com/

Batteries across your cell membranes power Life



Batteries across your cell membranes power Life





Energy lost by letting one positive ion go through membrane:

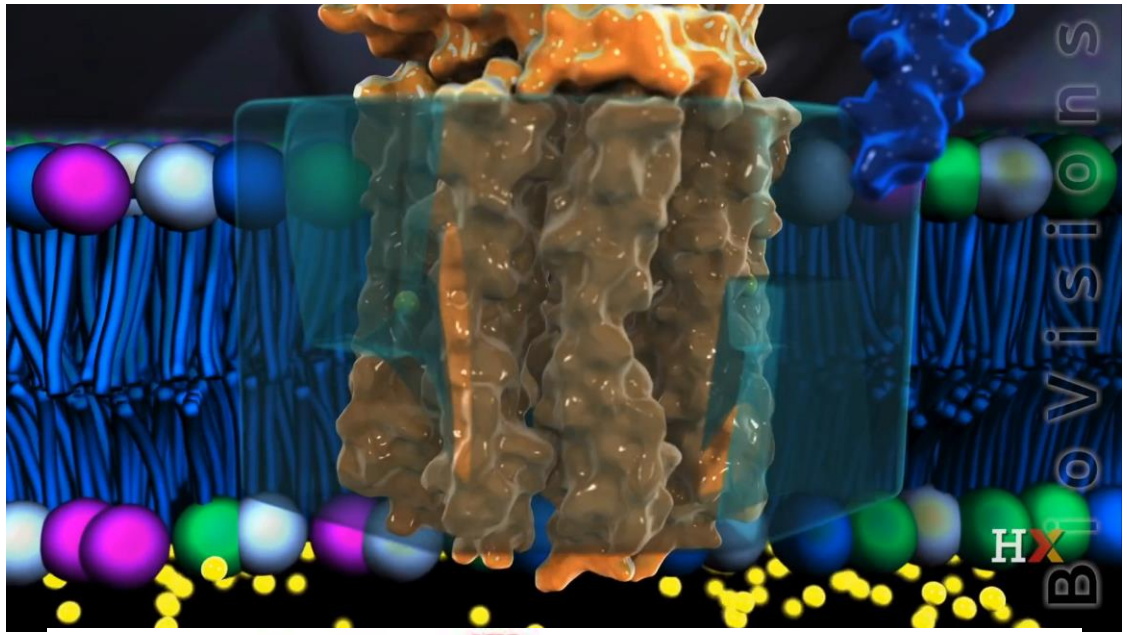
$E = qV = 0.1$ eV

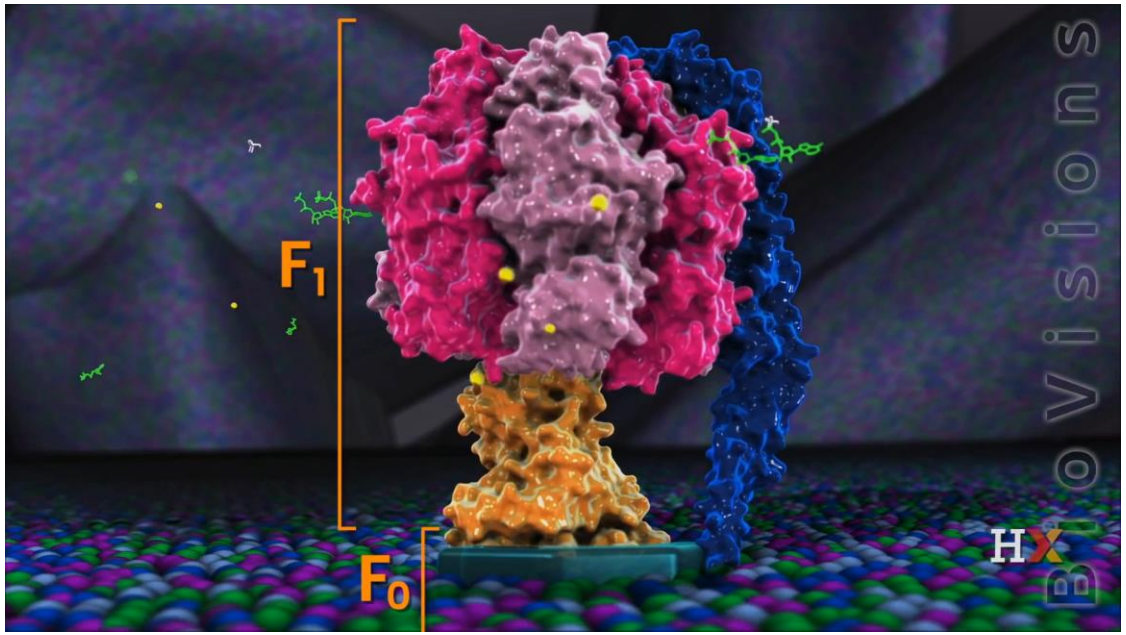
$k_BT = 1/40$ eV $= 0.025$ eV

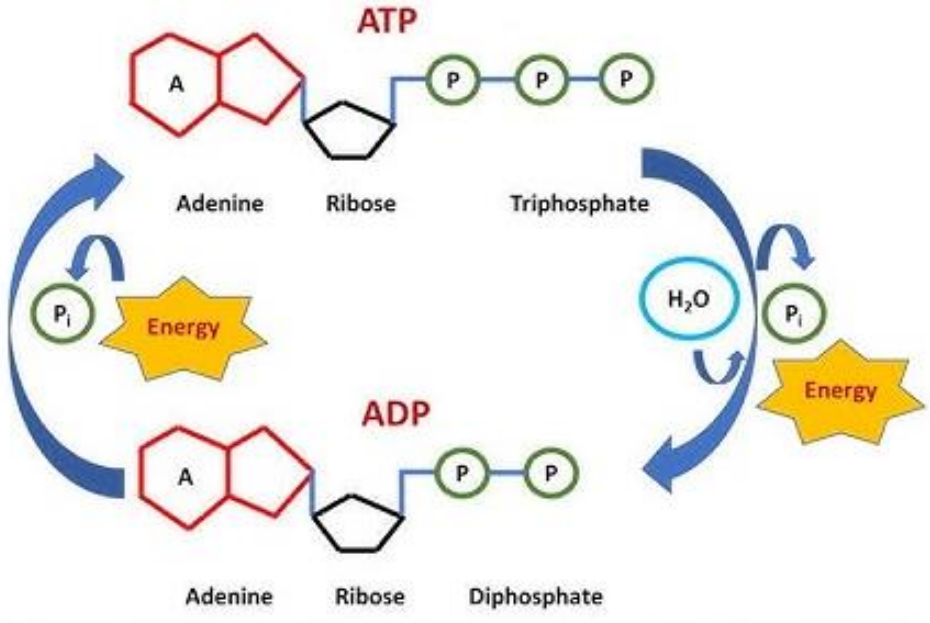
Therefore 4 k_BT of energy is generated every time a positive ion passes

3 k_BT energy is provided from entropy, making a total energy count as 7 k_BT

Membrane protein example







It requires 20-25 k_BT to convert ADP to ATP!